Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Data scientist and analytics engineer with **8+ years building ML pipelines and analytical frameworks on regulated financial data**. Built **Isolation Forest anomaly detection** at GMU - automated monitoring framework that surfaced anomalies manual review missed. Built **5-metric AI/ML evaluation engine (AgentProbe)** measuring false positive/negative rates across automated decisioning - the exact capability Chime's Dispute Experience team requires. Expert in **advanced SQL (window functions, CTEs)**, Python, and R. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

Anomaly Detection | AI/ML System Evaluation | Advanced SQL | Python | R | Root-Cause Analysis | KPI Frameworks | Experiment Design | Causal Inference | Tableau / Power BI | Financial Data Pipelines | Statistical Modeling

EXPERIENCE

George Mason University
Aug 2024 - May 2026

Research Associate - Analytics, Anomaly Detection & ML Modeling

Fairfax, Virginia

- Built **Isolation Forest anomaly monitoring framework** across 3 research pipelines - identified emerging anomaly patterns that manual expert review missed, **cut manual QA time by 70%**.
- Built regression models (**R^2 = 0.977**) in Python and R across WHO + UK Road Safety datasets spanning **190+ countries** - full lifecycle: data collection, feature engineering, validation, root-cause narrative, and stakeholder reporting.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - distilled complex ML findings into actionable insights, improving comprehension across two cohorts by **40%**.

Entrata India (Db Xento Systems)
Oct 2018 - Aug 2024

Software Engineer - Financial Data Analytics & Platform

Pune, India

- Processed **1.6M financial records** through Python and AI transformation pipelines - production data analytics on regulated financial datasets with output validation, quality gates, and compliance checkpoints.
- Tuned **200+ SQL queries** (window functions, CTEs, index optimization) - **30% production performance gain** across all client environments on complex financial data schemas.
- Achieved **100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines - **zero security incidents**; built 5+ analytical tools cutting resolution time by 40%.
- Eliminated **80% of manual operational workload** via Python/PL-SQL automation across 120+ Agile sprints - **100K+ MAU, 99.5% uptime** sustained over 6+ months.
- Conducted 300+ code reviews enforcing data quality, TDD/BDD, and secure coding standards; mentored 5+ engineers.

Entrata India (Db Xento Systems)
Jul 2017 - Sep 2018

Associate Software Engineer

Pune, India

- Improved system stability by **35%** building 20+ full-stack data modules (Python, PostgreSQL) - 99.5% uptime and 25% faster report generation.

PROJECTS

AgentProbe - AI/ML Evaluation Engine & False Positive/Negative Analysis
In Progress

2024-Present | Python, LangChain, LangGraph, FastAPI, ChromaDB | github.com/venkatalakshmiparimala/AgentProbe

- Built **5-metric evaluation engine** measuring false positive/negative rates across AI-enabled decisioning systems (factual accuracy, hallucination, retrieval precision, completeness, compliance) - **CI/CD integration blocking failing automated decisions before production**.
- Delivered end-to-end **model quality monitoring** via React dashboard + ChromaDB tracking regression drift and decision accuracy per run across concurrent test suites.

Global Health & Road Safety - Anomaly Detection & Root-Cause Analytics

2024-2025 | Python, R, Isolation Forest, scikit-learn, Tableau, ggplot2, Quarto | George Mason University

- Built **Isolation Forest anomaly detection framework** - identified emerging data anomalies across 3 active research pipelines and produced **root-cause narratives** for faculty leadership; cut manual review time by **70%**.

CrashClarity - Large-Scale Pattern Analysis & Stakeholder Reporting (104K Records)

2025 | Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering | Live: Portfolio

- Achieved **85%+ accuracy** on 104K+ records by building a Random Forest + Logistic Regression pipeline with experiment design and model evaluation (precision/recall/F1) - deployed interactive dashboard for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University — 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, causal inference, experiment design, anomaly detection, ML systems

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering — 2013-2017

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash

Analytics & ML: Isolation Forest, Random Forest, XGBoost, Logistic Regression, scikit-learn, Statistical Modeling, Causal Inference, A/B Testing, Anomaly Detection, Feature Engineering, NumPy, Pandas

Data & Infra: PostgreSQL, MySQL, AWS (S3, EC2, RDS, Glue), Docker, FastAPI, Flask, RabbitMQ, Microservices

Visualization: Tableau, Power BI, ggplot2, Plotly, Quarto; Looker/Mode (familiar)

AI & ML Eval: LangChain, LangGraph, OpenAI API, Anthropic Claude API, LLM-as-Judge Evaluation, CI/CD Eval Gates, ChromaDB, BERT, spaCy

Process: TDD/BDD, Git, JIRA, Agile/Scrum, PHI/HIPAA compliance, secure coding, audit logging